Topics in Macroeconomics

Academic Year: 2026/27 **Term:** 2nd Trimester **ECTS:** 6

INSTRUCTOR(S)

Lukas Boehnert

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BIOGRAPHY

Lukas Boehnert is a Research Economist at the Bank of Portugal, and an Academic Visitor at the University of Oxford. His primary research interests are in international macroeconomics and macro-development. He received his PhD in Economics from the University of Oxford in 2026.

He has held research positions at several policy institutions, including as a Summer Research Graduate at the ECB, a visiting student at the Minneapolis Fed, and as a PhD intern at the International Monetary Fund and the Swedish Riksbank. Before commencing his PhD, he has worked at the United Nation Development Program in New York. You can find further details on his website: www.lukasboehnert.com.

COURSE OVERVIEW

This graduate-level course introduces PhD students to key questions, methods, and recent research in macroeconomics, with a particular focus on empirical macroeconomics, international macroeconomics, and macro-development. The course emphasizes the interaction between theoretical models, empirical identification, and quantitative implementation. Its goal is to provide students with a broad understanding of current research frontiers while helping them develop the tools and perspective needed to formulate potential PhD research projects in these areas.

Part I: Empirical methods for estimation and identification in macroeconomics

The first part of the course focuses on empirical methods commonly used in modern macroeconomic research. We begin by studying vector autoregressions and local projections, emphasizing their conceptual differences, practical implementation, and respective advantages and limitations. We then turn to shift-share research designs and their applications in macroeconomic contexts. Throughout this part of the course, students will apply these methods in coding sessions that replicate influential empirical findings from the literature.

Part II: International Macroeconomics

The second part of the course provides an overview of classical and recent contributions in international macroeconomics and international finance. Topics include financial crises, monetary policy in open economies, exchange-rate dynamics, and capital flows. We will study the theoretical foundations of baseline open-economy models and examine how these models

are used to interpret empirical patterns and policy trade-offs. As in Part I, coding sessions will be used to replicate and extend selected results from the literature.

Part III: Macro-Development

The final part of the course examines recent research in macro-development, broadly defined. Topics include aggregate productivity, misallocation, regional growth, and spatial economics. We begin with classical misallocation frameworks and their implications for productivity and development. We then discuss recent advances in spatial economics that allow researchers to study regional dispersion, local labor markets, and the geographic dimensions of misallocation. This part of the course highlights how macroeconomic tools can be used to study development questions across countries and regions.

LEARNING OBJECTIVES

By the end of this module, students should be able to:

- Critically evaluate empirical macroeconomics papers, including their identification strategies, underlying assumptions, empirical implementation, and interpretation of results.
- Understand, design, implement, and interpret vector autoregressions, local projections, and shift-share research designs in macroeconomic contexts.
- Assess the strengths and limitations of classical and recent macroeconomic frameworks in international macroeconomics, international finance, and macro-development.
- Solve and quantitatively analyze smaller-scale versions of international macroeconomic models and misallocation frameworks using Matlab or Python.
- Replicate selected empirical and quantitative findings from the literature using appropriate coding, data-handling, and numerical methods.
- Apply the methods and models covered in the course to formulate, refine, and evaluate a research question relevant to their own PhD work.

TEACHING AND LEARNING METHODOLOGY

Sessions are structured around academic research papers, with an expectation of active and informed participation. Students are expected to engage with the assigned readings in advance. Lecture notes and materials are made available beforehand to ensure that class time is devoted to the critical assessment of the literature through discussion.

In addition, students will discuss one research paper of their choice and work on a written assignment applying the methods and knowledge from one part of the course, with the aim of producing work that could serve as an initial draft of a PhD thesis chapter.

REQUIRED BACKGROUND

Students are expected to have successfully completed Macroeconomics I and II and to possess graduate-level knowledge in econometrics, as the course relies on the development and estimation of empirical and structural models.

ASSESSMENT

Class Participation	individual grade	20%
Class Project	individual grade	50%
Paper discussion	individual grade	30%

Class participation is assessed on the quality of engagement with the readings, including preparation, critical evaluation, and informed contributions to discussions.

COURSE CONTENT

Part I: Empirical methods for estimation and identification in macroeconomics

1. Local Projection vs VARs: Comparison, Drawbacks and Implementation
2. Shift-share Instrumental Variables

Part II: International Macroeconomics

1. Financial crises and policy response in Open Economies
2. Monetary Policy in Open Economies

Part III: Macro-Development

1. Misallocation and Growth
2. Introduction to Spatial Economics

BIBLIOGRAPHY

Required Readings:

1. Empirical methods for estimation and identification in macroeconomics

   Jordà, Ò. (2023). Local projections for applied economics. *Annual Review of Economics*, *15*, 607-631.

   Olea, J. L. M., Plagborg-Møller, M., Qian, E., & Wolf, C. K. (2025). *Local projections or VARs? a primer for macroeconomists* (No. w33871). National Bureau of Economic Research.

   Plagborg-Møller, M., & Wolf, C. K. (2021). Local projections and VARs estimate the same impulse responses. *Econometrica*, *89*(2), 955-980.

   Montiel Olea, J. L., Plagborg-Møller, M., & Qian, E. (2022, May). Svar identification from higher moments: Has the simultaneous causality problem been solved?. In *AEA Papers and Proceedings* (Vol. 112, pp. 481-485). 2014 Broadway, Suite 305, Nashville, TN 37203: American Economic Association.

   Borusyak, K., Hull, P., & Jaravel, X. (2025). A practical guide to shift-share instruments. *Journal of Economic Perspectives*, *39*(1), 181-204.

   Borusyak, K., Jaravel, X., & Spiess, J. (2024). Revisiting event-study designs: robust and efficient estimation. *Review of Economic Studies*, *91*(6), 3253-3285.

   Code repositories:

   https://sites.google.com/site/oscarjorda/home/local-projections

   https://sites.google.com/view/borusyak/code

2. International Macroeconomics

   Mendoza, E. G. (2010). Sudden stops, financial crises, and leverage. *American Economic Review*, *100*(5), 1941-1966.

   Bianchi, J. and E. Mendoza (2018). "Optimal Time-Consistent Macroprudential Policy," Journal of Politicaly Economy 126, pp. 588-634.

   Clarida, R., Gali, J. and M. Gertler (2001). "Optimal Monetary Policy in Open Versus Closed Economies: An Integrated Approach," American Economic Review P&P, pp. 248-252.

Clarida, R., Gali, J. and M. Gertler (2001). "A Simple Framework for International Monetary Policy Analysis," Journal of Monetary Economics 49, pp. 879-904.

Gali, J. and T. Monacelli (2005). "Monetary Policy and Exchange Rate Variability in a Small Open Economy," Review of Economic Studies 72, pp. 707-734.

Benigno, G. and P. Benigno (2006). "Designing Targeting Rules for International Monetary Policy Cooperation," Journal of Monetary Economics 53, pp. 473-506.

Ferrero, A., Gertler, M. and Svensson, L. (2008), "Current Account Dynamics and Monetary Policy,", in Gali, J. and Gertler, M. (eds.): International Dimensions of Monetary Policy. University of Chicago Press, Chicago, IL.

Benigno, P. (2004). "Optimal Monetary Policy in a Currency Area," Journal of International Economics 63, pp. 293-320.

Gali, J. and T. Monacelli (2008). "Optimal Monetary and Fiscal Policy in a Currency Union," Journal of International Economics 76, pp. 116-132.

Ferrero, A. (2009). "Fiscal and Monetary Rules for a Currency Union," Journal of International Economics 77, pp. 1-10.

Eggertsson, G., Ferrero, A. and A. Raffo (2014). "Can Structural Reforms Help Europe?" Journal of Monetary Economics 61, pp. 2-22.

3. Macro-Development

Hsieh, Chang-Tai, and Peter J. Klenow. "The Life Cycle of Plants in India and Mexico" The Quarterly journal of economics 124.4 (2014)

Bils, Mark, Peter J. Klenow, and Cian Ruane. "Misallocation or mismeasurement?." Journal of Monetary Economics 124 (2021): S39-S56.

Restuccia, Diego, and Richard Rogerson. "Policy distortions and aggregate productivity with heterogeneous establishments." Review of Economic dynamics 11.4 (2008): 707-720.

Hsieh, Chang-Tai, and Peter J. Klenow. "Misallocation and manufacturing TFP in China and India." The Quarterly journal of economics 124.4 (2009): 1403-1448.

Krugman, P. (1991). Increasing returns and economic geography. *Journal of political economy*, *99*(3), 483-499.

Eaton, J., & Kortum, S. (2002). Technology, geography, and trade. *Econometrica*, *70*(5), 1741-1779.

Redding, S. J. (2016). Goods trade, factor mobility and welfare. *Journal of International Economics*, *101*, 148-167.

Redding, S. J., & Sturm, D. M. (2008). The costs of remoteness: Evidence from German division and reunification. *American Economic Review*, *98*(5), 1766-1797.

Durlauf, S., & Blume, L. E. (2016). *The new Palgrave dictionary of economics*. Springer.

Nagy, D. K. (2022). Quantitative economic geography meets history: Questions, answers and challenges. *Regional Science and Urban Economics*, *94*, 103675.

Allen, T., & Arkolakis, C. (2014). Trade and the Topography of the Spatial Economy. *The Quarterly Journal of Economics*, *129*(3), 1085-1140.

Eckert, F., & Peters, M. (2022). *Spatial structural change* (No. w30489). National Bureau of Economic Research.

Glaeser, E. L., Kallal, H. D., Scheinkman, J. A., & Shleifer, A. (1992). Growth in cities. *Journal of political economy*, *100*(6), 1126-1152.

ADDITIONAL RESOURCES

AI Support Tools: Through IAedu FCT, we offer our students access to enhanced versions of ChatGPT (5.2), DeepSeek (3.2), Claude Opus (4.5) and Llama (4 Maverick) with possibility to create AI Agents. Log in here with your CLSBE credentials.

CODE OF CONDUCT AND ETHICS

Católica Lisbon School of Business and Economics is a community of individuals with diverse backgrounds and interests who share certain fundamental goals. A crucial element to achieve these goals is the creation and maintenance of an atmosphere contributing to learning and personal growth for everyone in the community. The success of CATÓLICA-LISBON in attaining its goals and in maintaining its reputation of academic excellence depends on the willingness of its members, both collectively and individually, to meet their responsibilities.

Along with all the other members of our community, students are expected to follow professional standards and CATÓLICA-LISBON standards of Academic Integrity. Some details should be mentioned here: Please arrive on time for class with uninterrupted attendance for the duration of the class. Signing attendance sheet for anyone else in the class constitutes fraud and a violation of the CLSBE code of conduct. Use of computers and other electronic devices during the class is not allowed unless expressly requested by the instructor of the course. Students who persistently act in a disruptive and disrespectful manner during the class session may be invited to leave.

Students are expected to behave at all times according to the fundamental principles of academic integrity, including honesty, trust, fairness, respect, and responsibility. In particular,

a. In **individual graded assignments** of any type, students may not collaborate with others or use any materials without explicit permission from the instructor of the course;

b. In **group assignments** and reports, all students listed as authors should have performed a substantial amount of work for that assignment;

c. It is dishonest to fabricate or falsify data in experiments, surveys, papers, reports or other circumstances; fabricate source material in a bibliography or "works cited" list; or provide false information in other documents in connection with academic efforts;

d. **Plagiarizing**, i.e. "to steal and pass off the ideas or words of another as one's own and or to use another's production without crediting the source" (Merriam-Webster Dictionary) is an Academic Integrity breach. It can be avoided by using proper methods of documentation and acknowledgement. Visit this guide for additional resources on how to avoid plagiarism in your written submissions: https://www.turnitin.com/papers/understanding-the-turnitin-similarity-report-student-guide

e. In **exams** students must not receive or provide any unauthorized assistance. During an examination, students may use only material and items authorized by the faculty. Use of smartwatches or other communication devices is not permitted during the exam.

Academic integrity breaches will be dealt with in accordance with the school's code of Academic Integrity: https://www.clsbe.lisboa.ucp.pt/system/files/assets/files/academicintegritycode.pdf